

BELLE RIVER, ON, Canada

WMO: 716363

Lat: 42.300N

Lon: 82.700W

Elev: 184

StdP: 99.14

Time Zone: -5.00 (NAE)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-12.2	-18.2	0.8	-13.6	-15.8	1.0	-11.3	14.1	-5.0	13.0	-3.1	5.2	240	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	31.6	24.2	30.0	23.7	28.5	22.8	25.9	29.6	24.9	28.4	23.9	27.0	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.4	28.3	23.8	19.1	27.0	23.0	18.1	25.9	81.3	29.7	76.8	28.5	72.8	27.1	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.3	9.9	DB	-16.7	35.1	3.4	2.0	-19.1	36.6	-21.1	37.7	-23.1	38.9	-25.5	40.3	(4)
(5)			WB	-17.0	27.9	3.4	1.3	-19.4	28.8	-21.4	29.6	-23.3	30.4	-25.7	31.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-3.4	-1.2	2.7	9.4	15.0	20.9	23.3	22.0	18.8	12.0	6.0	0.0	(6)	
	DBStd	10.24	5.74	4.71	5.46	4.68	4.37	4.17	2.80	2.66	3.84	4.24	4.49	5.31	(7)	
	HDD10.0	1514	416	315	237	65	6	0	0	0	1	27	135	313	(8)	
	HDD18.3	3307	673	548	486	272	123	22	1	3	41	201	369	568	(9)	
	CDD10.0	1701	0	1	9	46	162	327	412	373	264	88	16	3	(10)	
	CDD18.3	454	0	0	1	3	21	99	154	118	54	4	0	0	(11)	
	CDH23.3	2851	0	0	2	30	146	738	1047	590	280	17	0	0	(12)	
	CDH26.7	795	0	0	0	3	33	252	317	135	53	1	0	0	(13)	
(14) Wind	WSAvg	4.2	4.9	4.6	5.0	5.1	3.9	3.3	3.4	3.6	3.7	4.0	4.7	4.6	(14)	
(15) Precipitation	PrecAvg	895	63	58	66	83	88	80	87	81	89	65	68	69	(15)	
	PrecMax	1413	105	135	142	140	170	134	166	167	185	146	177	138	(16)	
	PrecMin	710	21	17	16	31	32	30	26	11	26	10	16	31	(17)	
	PrecStd	143	24	28	29	29	42	33	36	38	48	31	32	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	15.9	22.5	26.9	30.1	33.5	34.2	32.2	30.6	25.4	18.5	16.3	(19)	
		MCWB	12.7	13.8	16.5	19.6	21.3	25.2	26.2	24.8	22.4	19.9	14.8	14.1	(20)	
	2%	DB	10.6	10.0	17.7	23.2	27.3	31.5	31.8	29.8	28.1	22.9	16.4	12.6	(21)	
		MCWB	10.1	8.1	13.7	16.9	21.3	23.7	25.0	24.4	22.1	17.6	14.3	11.4	(22)	
	5%	DB	6.9	6.9	14.1	19.6	24.6	29.8	29.9	28.0	26.5	20.7	14.7	9.0	(23)	
		MCWB	5.9	5.6	11.5	14.8	18.8	23.2	24.1	23.2	21.2	16.6	13.0	7.8	(24)	
	10%	DB	3.4	4.4	10.4	16.5	22.0	27.6	28.3	26.4	24.7	18.4	12.6	6.4	(25)	
		MCWB	2.7	3.2	8.1	12.4	18.1	22.1	22.9	22.3	20.6	15.5	10.7	5.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	13.9	16.9	20.6	23.8	26.6	28.2	26.7	24.6	21.5	16.4	15.3	(27)	
		MCDB	13.2	15.2	21.8	25.8	27.2	31.7	32.6	30.2	27.7	24.5	17.9	15.9	(28)	
	2%	WB	10.4	8.3	14.1	17.7	21.9	25.2	26.2	25.3	23.5	19.1	14.8	11.4	(29)	
		MCDB	10.5	9.6	16.8	21.3	25.5	29.3	29.6	28.7	26.4	21.5	16.1	12.5	(30)	
	5%	WB	5.6	5.7	11.5	15.5	20.2	24.1	25.2	24.1	22.5	17.4	13.1	8.1	(31)	
		MCDB	6.8	6.8	14.1	18.9	23.1	28.0	28.6	26.6	25.1	19.7	14.4	9.0	(32)	
	10%	WB	2.7	3.4	8.0	13.0	18.5	23.0	24.1	23.1	21.3	15.9	11.2	5.4	(33)	
		MCDB	3.4	4.2	10.1	15.5	21.3	26.7	27.2	25.5	23.6	18.0	12.3	6.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.3	7.3	8.2	8.7	8.6	8.3	7.4	8.0	7.1	5.8	5.2	(35)	
		MCDBR	8.9	9.5	13.0	13.5	12.2	10.9	10.5	9.8	10.4	10.5	8.3	7.8	(36)	
	5% WB	MCWBR	8.3	8.0	9.4	8.8	7.1	5.2	4.8	5.2	5.3	6.1	7.1	6.7	(37)	
		MCWBR	8.8	9.4	12.6	12.1	11.0	9.8	9.7	8.7	9.0	8.9	7.5	7.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.303	0.332	0.353	0.397	0.430	0.445	0.435	0.425	0.386	0.359	0.320	0.311	(40)	
		taud	2.398	2.292	2.336	2.249	2.207	2.209	2.262	2.279	2.363	2.421	2.492	2.429	(41)	
	Ebn at Noon		838	862	889	870	848	833	837	834	844	825	816	795	(42)	
		Edh at Noon	79	103	112	132	141	141	133	127	108	91	73	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.41	2.29	3.48	4.63	5.44	6.14	6.11	5.31	4.28	2.78	1.72	1.21	(44)	
	RadStd		0.17	0.25	0.34	0.40	0.46	0.38	0.31	0.21	0.36	0.23	0.22	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page